Installing Jenkins on Ubuntu Server

Introduction

Jenkins is an open-source automation server used to automate software build, testing, and deployment processes. Before using Jenkins, it must be installed and configured on a server.

This document explains how to install Jenkins on an Ubuntu Server.

Prerequisites

Before installing Jenkins, ensure that:

- Ubuntu Server is running.
- You have a user account with sudo privileges.
- Internet connectivity is available.
- Port 8080 is allowed through the firewall (if applicable).

Step 1: Update the System

Before installing any software, update the package repository and upgrade existing packages.

Run the following command:

```
sudo apt update && sudo apt upgrade -y
```

Explanation

- `apt update` downloads the latest package information.
- `apt upgrade` installs the latest available package updates.
- `-y` automatically answers “yes” to prompts.

Step 2: Install Java

Why Java is Required

Jenkins is a Java-based application and requires Java Runtime Environment (JRE) or Java Development Kit (JDK) to run.

Jenkins currently supports OpenJDK 21 and later versions.

Install OpenJDK 21:
sudo apt install openjdk-21-jdk -y

Verify Java Installation

Check the installed Java version:

```
java -version
```

Example Output

```
openjdk version "17.0.x"
OpenJDK Runtime Environment
OpenJDK 64-Bit Server VM
```

If Java version information is displayed, Java has been installed successfully.

Step 3: Add Jenkins Repository and Install Jenkins

Ubuntu's default repositories may not contain the latest Jenkins version.

Therefore, add the official Jenkins repository.

Add Jenkins Repository Key

Run:

```
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | \
sudo gpg --dearmor -o /usr/share/keyrings/jenkins-keyring.gpg
```

Purpose

This command downloads and stores Jenkins' GPG key, which is used to verify package authenticity.

Add Jenkins Repository

Run:

```
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.gpg] \
https://pkg.jenkins.io/debian-stable binary/" | \
sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

Purpose

This command adds the official Jenkins package repository to Ubuntu.

Update Package Information

After adding the repository, refresh package information:

```
sudo apt update
```

Install Jenkins

Install Jenkins using:

```
sudo apt install -y jenkins
```

What Happens?

Ubuntu will:

- Download Jenkins packages.
- Install Jenkins.
- Create Jenkins service files.
- Create the Jenkins home directory.

Default Jenkins Home Directory:

```
/var/lib/jenkins
```

Step 4: Start and Enable Jenkins Service

Start Jenkins

Run:

```
sudo systemctl start jenkins
```

This starts the Jenkins service immediately.

Enable Jenkins at Boot

Run:

```
sudo systemctl enable jenkins
```

This ensures Jenkins starts automatically whenever the server reboots.

Verify Jenkins Service Status

Check whether Jenkins is running:

```
sudo systemctl status jenkins
```

Expected Output

```
Active: active (running)
```

If you see “active (running)”, Jenkins is running successfully.

Step 5: Configure Firewall (Optional)

If UFW (Uncomplicated Firewall) is enabled, allow Jenkins traffic.

```
sudo ufw allow 8080
sudo ufw reload
```

Verify:

```
sudo ufw status
```

Port 8080 should be listed as allowed.

Step 6: Access Jenkins Web Interface

Open a browser and navigate to:

```
http://<server-ip>:8080
```

Example:

```
http://192.168.1.100:8080
```

You should see the Jenkins Unlock screen.

Step 7: Retrieve Initial Admin Password

For security reasons, Jenkins generates a temporary administrator password during installation.

Retrieve the password using:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Example Output

```
8c56d0f57c4d4d6bb3c2d8dcb95a1a9f
```

Copy the displayed password.

Step 8: Unlock Jenkins

1. Open the Jenkins web page.
2. Paste the initial admin password.
3. Click **Continue**.

Step 9: Install Recommended Plugins

Jenkins will display plugin installation options.

Select:

```
Install Suggested Plugins
```

Common Plugins Installed

- Git Plugin
- Pipeline Plugin
- Docker Plugin
- Credentials Plugin
- SSH Plugin
- Maven Integration Plugin

These plugins provide essential Jenkins functionality.

Step 10: Create Administrator User

After plugin installation:

1. Enter username.
2. Enter password.
3. Enter full name.
4. Enter email address.

Click:

```
Save and Continue
```

This creates the first Jenkins administrator account.

Step 11: Configure Jenkins URL

Jenkins asks for the URL used to access the server.

Example:

```
http://192.168.1.100:8080
```

or

```
https://jenkins.company.com
```

Click:

```
Save and Finish
```

Jenkins Installation Verification

After successful login:

You should see the Jenkins Dashboard.

Example dashboard sections:

- New Item
- Build History
- Manage Jenkins
- Manage Plugins
- Nodes
- Credentials

At this stage Jenkins is fully installed and ready for creating jobs and pipelines.

Important Jenkins Directories

Directory	Purpose
/var/lib/jenkins	Jenkins Home Directory
/var/log/jenkins	Jenkins Logs
/etc/default/jenkins	Jenkins Configuration
/var/cache/jenkins	Jenkins Cache Files

Useful Jenkins Commands

Start Jenkins

```
sudo systemctl start jenkins
```

Stop Jenkins

```
sudo systemctl stop jenkins
```

Restart Jenkins

```
sudo systemctl restart jenkins
```

Check Status

```
sudo systemctl status jenkins
```

View Logs

```
sudo journalctl -u jenkins -f
```

Installation Workflow Summary

```
Update Ubuntu
        ↓
Install Java
        ↓
Add Jenkins Repository
        ↓
Install Jenkins
        ↓
Start Jenkins Service
        ↓
Access Jenkins Web UI
        ↓
Unlock Jenkins
        ↓
Install Plugins
        ↓
Create Admin User
        ↓
Jenkins Ready for Use
```

Conclusion

Jenkins installation on Ubuntu involves:

1. Updating the operating system.
2. Installing Java.
3. Adding the Jenkins repository.
4. Installing Jenkins.
5. Starting and enabling the Jenkins service.
6. Accessing the Jenkins web interface.
7. Installing plugins and creating an administrator account.

After completing these steps, Jenkins is ready to automate software build, testing, and deployment workflows.